Disk scheduling Algorithms manage how data is read from and written to a computer's hard disk.

The main goal is to determine in which disk read & write requests are processed.

→ Reduce the time it takes to access data & improve overall s/m efficiency.

Characteristics of Disk scheduling

Seek time → Time taken to move the disk arm to track where data is located.

Rotational latency – Time taken for the desired sector to rotate under the r/w head.

Transfer time → Time taken to actually read or write data

Disk Access time = Seek time + Rotational latency + transfer time.

Advantages of FCFS
→ Every request gets a fair chance
→ NO indefinite postponement.

Disadvantages
→ Does not try to optimize seek time
→ May not provide best possible service

SSTF
→ Advantages → Avg response time decreases
Throughput increases

Disadvantages
→ Overhead to calculate seek time in advance
→ can cause starvation for a request if it has a higher seek time as compared to incoming requests.
→ The high variance of response time as SSTF favor only some requests.

C-SCAN
Advantage → Eliminates starvation, Better performance.
→ Disk arm moves one direction only
→ After reaching the end, it jumps to the beginning
→ Services request in circular manner
Disadvantage → Extra head movement
→ Increase seek time
→ More overhead

~~LOOK & START~~

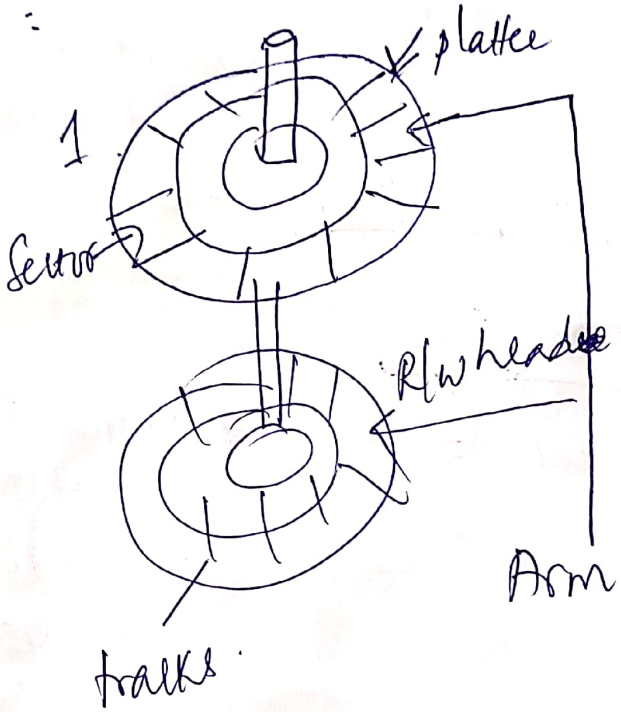
SCAN
Advantages → High throughput
→ Low variance & response time
→ Avg response time.
Disadvantages → Head may move to disk end unnecessarily
→ High waiting time for some requests
→ New requests may wait longer.

Disk Scheduling Algorithms.

- FCFS
- SSTF
- :
- :

Goal is to minimize the seek time.

seek time → time taken to reach up to desired track.



platter → surface → tracks → sector

Disk Access time = seek time + Rotational latency + transfer time

(waiting for sector to rotate under head) (Actual data transfer)

FCFS

A disk contains 200 tracks (0–199). Request Queue contains track no. 82, 170, 43, 140, 24, 16, 190 respectively. Current posn of R/w head = 50. Calculate total no. of tracks movement by R/w head.

0 16 24 43 50 82 140 170 190 199

Total no. of tracks movement = (82-50) + (170-82) + (170-43) + (140-43) + (140-24) + (24-16) + (190-16)

= 32 + 88 + 127 + 97 + 116 + 8 + 174

= 642

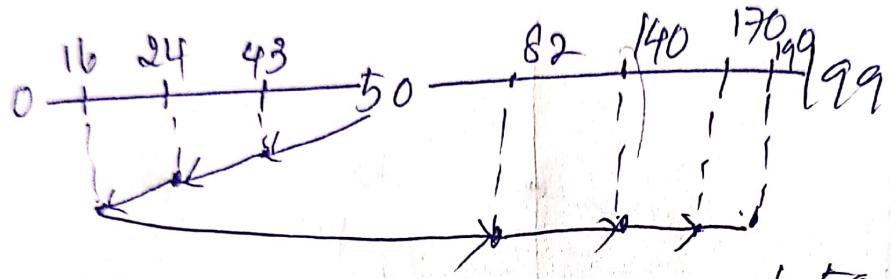
Disk

SSTF
[It gives optimal result]

For the above problem calculate total no. of tracks movement by R/W head using shortest seek time first?

If R/W head takes 1 ns to move from one track to another then total time taken = ?

(Need to go nearest)



208 x 1 ns
= 208 ns

Total No. of tracks movement = (50-43) + (43-24) + (24-16) + (82-16) + (140-82) + (170-140) + (190-170)

or (50-16) + (190-16)

can be done

14 + 174 = 208

problem is Starvation / overhead.

Disk scheduling

→ FCFS scheduling. (First come First served algorithm)

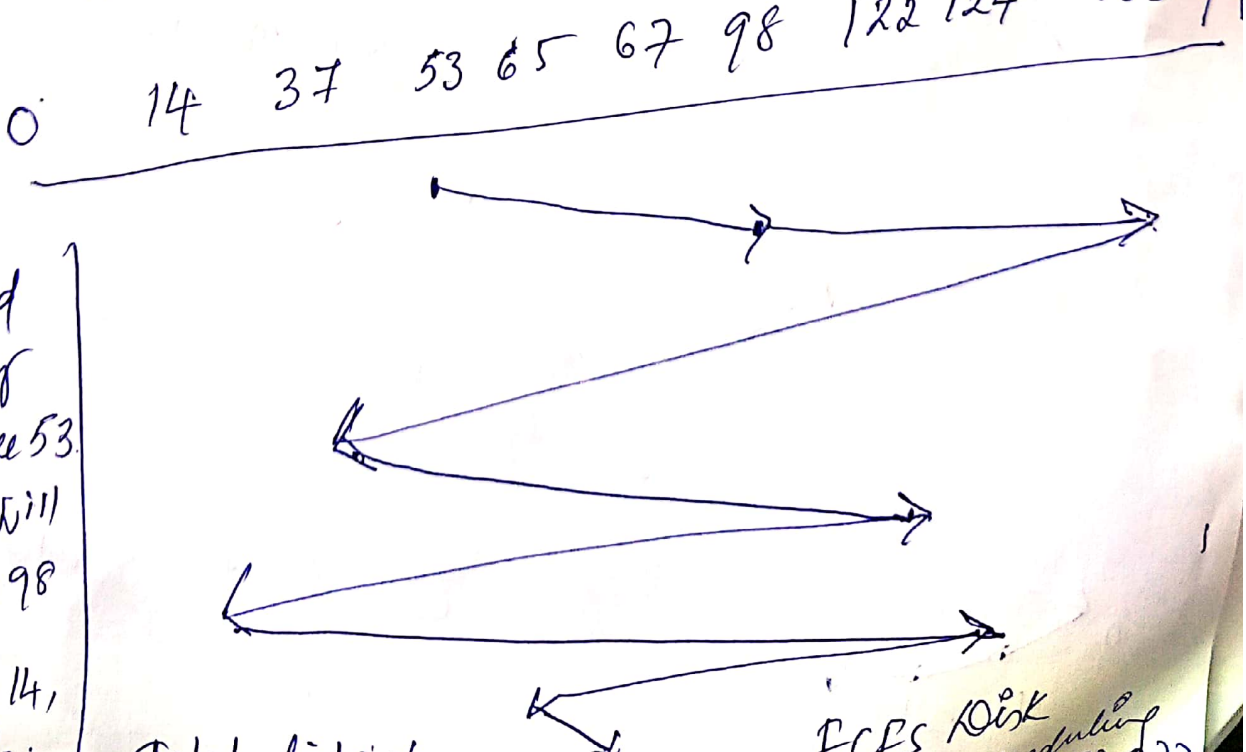
The simplest form of disk scheduling is, of course, the first come, first served (FCFS) algorithm. This algorithm is fair, but generally does not provide fastest service.

Ex:- consider disk queue with requests for I/O to blocks on cylinders

98, 183, 37, 122, 14, 124, 65, 67

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53.



Disk head is initially at cylinder 53. First it will from 53 to 98 then to 183, 37, 122, 14, 124 — — —

Total distance covered by disk arm = (98 − 53) + (183 − 98) + (183 − 37) + (122 − 14) + (124 − 65) + (67 − 65)

FCFS Disk Scheduling

3) Scan algorithm — consider above problem

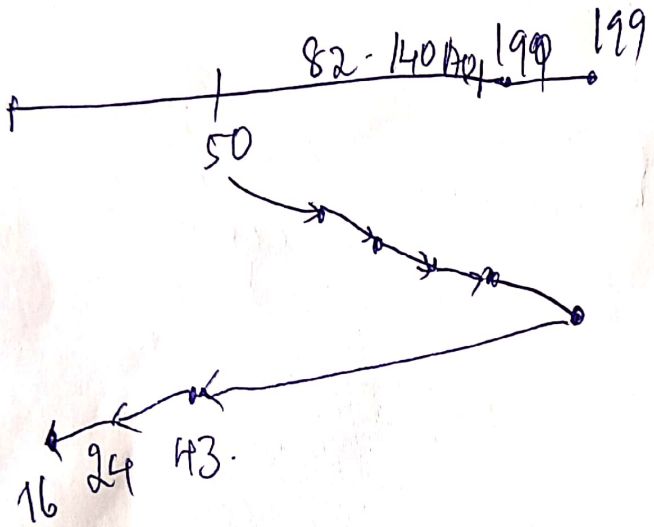
200 tracks (0 to 199)

[Move to (larger value) or smaller value given in Question]

Request Queue → 82, 170, 43, 140, 24, 16, 190

current posn of R/w head = 50

[Direction towards larger value]



[It will go to end ie 199.

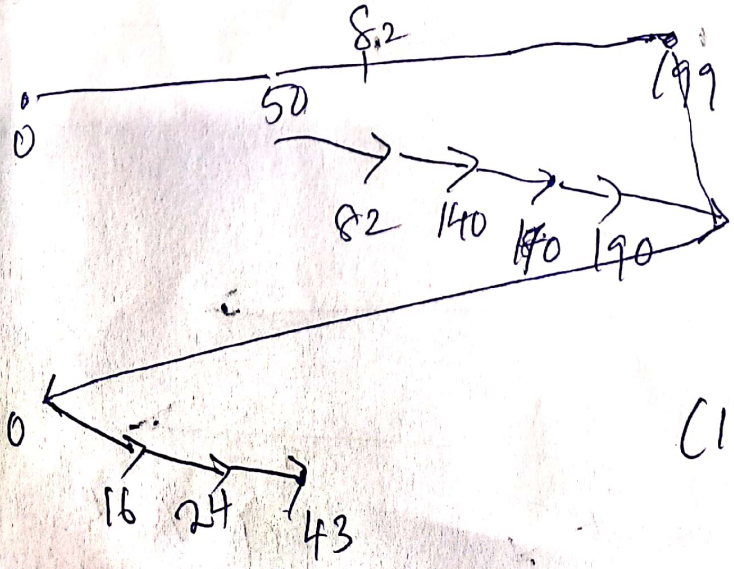
(199 − 50) + (199 − 16)

= 332

4) C-scan algorithm — consider above problem

↓ circular scan

→ 82, 170, 43, 140, 24, 16, 190.



R/w head posn = 50.

→ Direction towards large value.

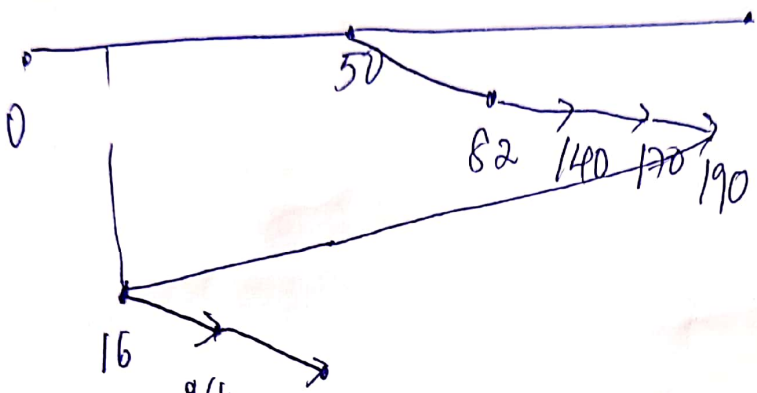
(199 − 50) + (199 − 0) + (43 − 0)

= 391

C LOOK Algorithm in Disk Scheduling

Consider same problem

82, 170, 43, 140, 24, 16, 190.



Direction towards large value.

[In C-look no seeking during retracing]

(190-50) + (190-16) + (24-16) + (43-24) + (43-16)

LOOK

Disk head looks ahead & goes only till the last request in that direction & then reverse. = 341

Selection of Disk scheduling Algo.

Given so many Algos how do you choose best algo.

SSTF increases performance over FCFS

SCAN & C-SCAN perform better but place a heavy load on the disk.

Types of Requests are also imp.